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Question **1**

Complete

Not graded

Does the infinite series $\sum_{n=1}^{\infty} \frac{cos^2 n}{n^2}$ converge or diverge?

Select one:

- a. Converges
- b. Diverges
- c. Cannot be determined

The series resembles the p-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ which converges because $p = 2 > 1$. In order for the new series to converge, the term of the p-series must dominate. Because $\cos^2 n \leq 1$, and $\frac{\cos^2 n}{n^2} \leq \frac{1}{n^2}$, as the numerator increases, the fraction increases. Therefore, the series $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n^2}$ converges by comparison with p-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

The correct answer is: Converges

Question **2**

Complete

Not graded

Does the following infinite series converge or diverge?

$$\sum_{n=1}^{\infty} \frac{1}{n2^n}$$

Select one:

- a. It converges.
- b. It diverges.
- c. It cannot be determined.

The series resembles the geometric series $\sum_{n=1}^{\infty} \frac{1}{2^n} = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$, which converges because $|r| = \frac{1}{2} < 1$. In order for the new series to converge, the term of the geometric series must dominate. Thus, $0 < \frac{1}{n2^n} \leq \frac{1}{2^n}$ as the denominator decreases, the fraction increases. Therefore, the series $\sum_{n=1}^{\infty} \frac{1}{n2^n}$ converges in comparison to the geometric series $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$.

The correct answer is: It converges.

Question **3**

Complete

Not graded

Does the following infinite series converge or diverge?

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+1}}$$

Select one:

- a. It converges.
- b. It diverges.
- c. It cannot be determined.

$\sum_{n=1}^{\infty} \frac{1}{\sqrt{2}} \frac{1}{n} = \frac{1}{\sqrt{2}} \sum_{n=1}^{\infty} \frac{1}{n}$ diverges because it is a harmonic series. In order for the new series to diverge, the term of the harmonic series must be smaller than the term of the new series. Because $\frac{1}{\sqrt{n^2+1}} > \frac{1}{\sqrt{n^2+n^2}} = \frac{1}{\sqrt{2n^2}} = \frac{1}{\sqrt{2}} \frac{1}{n} > 0,$ as the denominator increases, the fraction decreases. Therefore, the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+1}}$ diverges in comparison to $\frac{1}{\sqrt{2}} \sum_{n=1}^{\infty} \frac{1}{n}.$

The correct answer is: It diverges.

Question **4**

Complete

Not graded

Does the infinite series $\sum_{n=1}^{\infty} \frac{1}{n - \log n}$ converge or diverge?

Select one:

- a. Converges
- b. Diverges
- c. Cannot be determined

The series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges because it is a harmonic series. In order for the new series to diverge, the terms of the harmonic series must be smaller than the terms of the new series. You get $\frac{1}{n - \log n} > \frac{1}{n} > 0$ because n is always greater than $\log n$, and as the denominator increases, the fraction decreases. Therefore, $\sum_{n=1}^{\infty} \frac{1}{n - \log n}$ diverges by comparison with $\sum_{n=1}^{\infty} \frac{1}{n}$.

The correct answer is: Diverges

Question **5**

Complete

Not graded

Does the infinite series $\sum_{n=1}^{\infty} \frac{2 + \sin n}{n^2}$ converge or diverge?

Select one:

- a. Converges
- b. Diverges
- c. Cannot be determined

The series resembles the p-series $\sum_{n=1}^{\infty} \frac{1}{n^2}$, which converges because $p = 2 > 1$. In order for the new series to converge, the term of the p-series must dominate. Because $1 < 2 + \sin n < 3$, $0 < \frac{2 + \sin n}{n^2} < \frac{3}{n^2}$. As the numerator increases, the fraction increases. Therefore, the series $\sum_{n=1}^{\infty} \frac{2 + \sin n}{n^2}$ converges by comparison with the p-series $3\sum_{n=1}^{\infty} \frac{1}{n^2}$.

The correct answer is: Converges

Question **6**

Complete

Not graded

Evaluate the following as true or false.

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are infinite term series. Then $\sum_{n=1}^{\infty} a_n$ diverges if $\sum_{n=1}^{\infty} b_n$ diverges, and $a_n \geq b_n$ for all n.

Select one:

- True
- False

Look at the theorem carefully. The direct comparison test requires both the given series $\sum_{n=1}^{\infty} a_n$, where $a_n > 0$, and the familiar series $\sum_{n=1}^{\infty} b_n$, where $b_n > 0$.

The correct answer is 'False'.

Question **7**

Complete

Not graded

Does the following infinite series converge or diverge?

$$\sum_{n=1}^{\infty} \frac{1}{n+\sqrt{n}}$$

Select one:

- a. It converges.
- b. It diverges.
- c. It cannot be determined.

The series $\sum_{n=1}^{\infty} \frac{1}{2n} = \frac{1}{2}\sum_{n=1}^{\infty} \frac{1}{n}$ diverges because it is a harmonic series. In order for the new series to diverge, the term of the harmonic series must be smaller than the term of the new series. Because $\frac{1}{n+\sqrt{n}} > \frac{1}{n+n} = \frac{1}{2n} > 0$ (which is a result of $n > \sqrt{n}$), as the denominator increases, the fraction decreases. Therefore, $\sum_{n=1}^{\infty} \frac{1}{n+\sqrt{n}}$ diverges in comparison to $\frac{1}{2}\sum_{n=1}^{\infty} \frac{1}{n}$.

The correct answer is: It diverges.

Question **8**

Complete

Not graded

Does the following infinite series converge or diverge?

$$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n+n^2}$$

Select one:

- a. It converges.
- b. It diverges.
- c. It cannot be determined.

The series resembles the p-series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$,

which converges because $p = \frac{3}{2} > 1$.

In order for the new series to converge,

the term of the p-series must dominate.

Thus, $0 < \frac{\sqrt{n}}{n+n^2} < \frac{\sqrt{n}}{n^2} = \frac{1}{n^{3/2}}$ as the

denominator decreases, the fraction

increases. Therefore, the series

$\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n+n^2}$ converges in comparison to

the p-series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$.

The correct answer is: It converges.

Question **9**

Complete

Not graded

Does the infinite series $\sum_{n=1}^{\infty} \frac{1}{n!}$ converge or diverge?

Select one:

- a. Converges
- b. Diverges
- c. Cannot be determined

Notice that the denominator of the series $n!$ is greater than 2^{n-1} :

$n! = n \cdot (n-1) \cdot (n-2) \cdots 4 \cdot 3 \cdot 2 \cdot 1 >$

$2 \cdot 2 \cdot 2 \cdots \cdot 2 \cdot 2 \cdot 2 \cdot 1 = 2^{n-1}$

Therefore $\frac{1}{n!} < \frac{1}{2^{n-1}}$ for $n > 1$, as the denominator decreases, the fraction increases. The geometric series $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 2\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$, which converges because $|r| = \frac{1}{2} < 1$.

Therefore, the series $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges by comparison with the geometric series $2\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$.

The correct answer is: Converges

Question **10**

Complete

Not graded

Does the following infinite series converge or diverge?

$$\sum_{n=1}^{\infty} \frac{1}{n + n^{3/2}}$$

Select one:

- a. It converges.
- b. It diverges.
- c. It cannot be determined.

The series resembles the p-series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$, which converges because $p = \frac{3}{2} > 1$. In order for the new series to converge, the term of the p-series must dominate. Thus, $0 < \frac{1}{n + n^{3/2}} < \frac{1}{n^{3/2}}$ as the denominator decreases, the fraction increases.

Therefore, the series $\sum_{n=1}^{\infty} \frac{1}{n + n^{3/2}}$ converges in comparison to the p-series $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$.

The correct answer is: It converges.

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